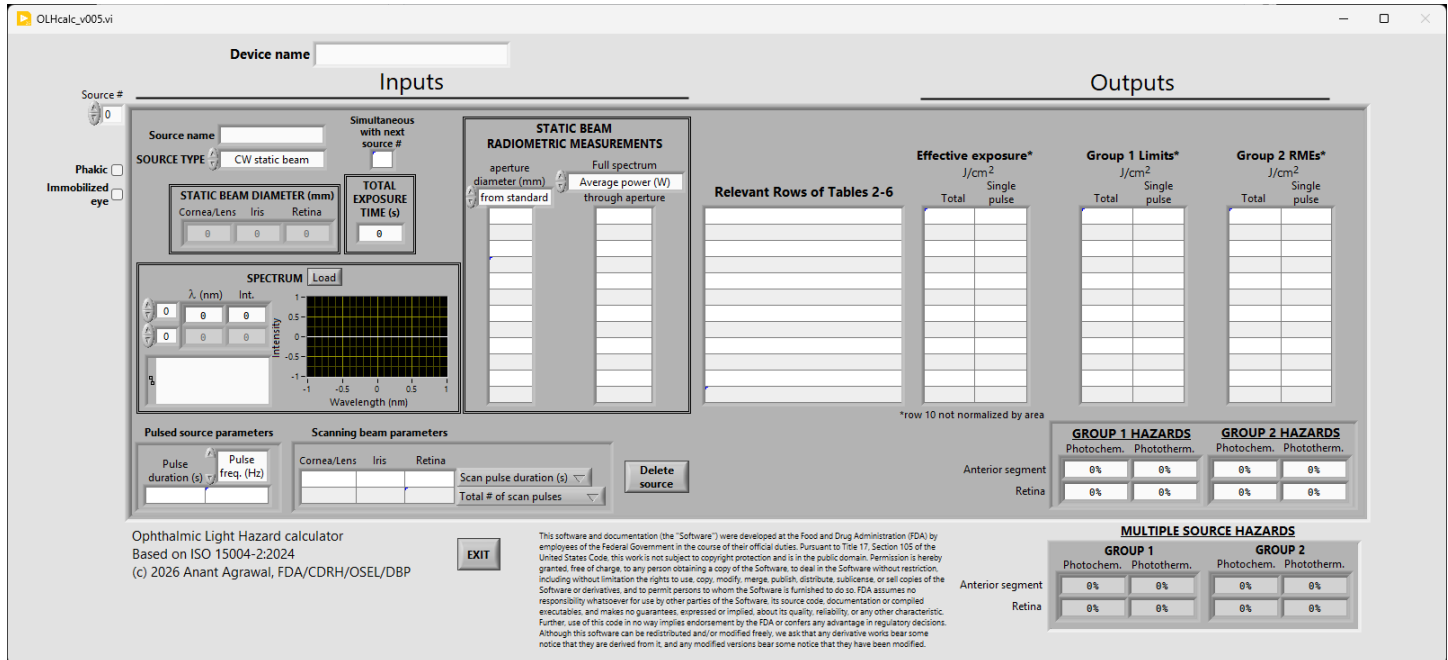


OLHcalc User Manual

Software version: 005

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The screenshot shows the OLHcalc app interface. It is divided into 'Inputs' and 'Outputs' sections. The 'Inputs' section includes fields for 'Device name', 'Source #', 'Source name', 'Source type' (set to 'CW static beam'), 'Simultaneous with next source #', 'Static beam diameter (mm)' for Cornea/Lens, Iris, and Retina, 'Total exposure time (s)', 'Spectrum' (with wavelength and intensity plots), 'Pulsed source parameters' (pulse duration and frequency), and 'Scanning beam parameters' (Cornea/Lens, Iris, Retina, scan pulse duration, and total number of scan pulses). The 'Outputs' section displays 'Effective exposure' (J/cm²) for Total and Single pulse, 'Group 1 Limits' (J/cm²) for Total and Single pulse, and 'Group 2 RMEs' (J/cm²) for Total and Single pulse. It also shows 'Relevant Rows of Tables 2-6' and 'GROUP 1 HAZARDS' and 'GROUP 2 HAZARDS' for Anterior segment and Retina. A 'DELETE source' button is present. At the bottom, there is a disclaimer and an 'EXIT' button.

Screenshot of OLHcalc app

1. Purpose

OLHcalc calculates **ophthalmic light hazards** based on definitions, logic, and formulas in **ISO 15004-2:2024 (Ophthalmic instruments — Fundamental requirements and test methods —Part 2: Light hazard protection)** for a device with one or more light sources irradiating the eye and computes combined hazards for sources that are active simultaneously.

Use this app to:

- enter source spectrum and exposure inputs,
- calculate per-source hazard results,
- and review multiple-source hazards.

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2. Assumptions

- The user is familiar with the ISO 15004-2:2024 document, particularly Tables 2-6.
- All scanning source beams have a circular or square cross-section.
- For multiple source devices, each beam irradiates the same portion of any particular ocular surface.

3. Inputs

Name	Required?	Usage
Device name	No	Device identifier for documentation purposes
SOURCE #	Yes	Array counter starting with zero for each source's page of inputs and outputs
Phakic	No	Check box to calculate effective exposures using phakic (less conservative) weighting functions. Leave unchecked for aphakic (more conservative) weighting functions.
Immobilized eye	No	Check box if eye is physically fixed or whose movements are compensated so that a retinal image does not move. This term also refers to a retinal image stabilized by eye tracking technology. It does not refer to an eye that is maintaining voluntary fixation, e.g. during an ophthalmic examination.
Source name	No	Source identifier for documentation purposes
SOURCE TYPE	Yes	Select one of three types: CW static beam, Pulsed static beam, CW scanning beam
Simultaneous with next source #	No	Enter a value only if this source emits simultaneously with one of the next sources in sequence for the device. The value must be the next Source # value in sequence. For example, if a device has three sources which all irradiate simultaneously, then for Source #0 enter a value of 1 and for Source #1 enter a value of 2, and nothing is entered for Source #2. If only the first and third sources are simultaneous, then 2 is entered for Source #0 and nothing is entered for Sources #1 and #2.
STATIC BEAM DIAMETER (mm)	Yes	Enter the beam diameter on each ocular surface. For static sources with noncircular beams, use the average of the maximum and minimum beam cross-section lengths. If an ocular surface is not irradiated, enter zero. A nonzero value must be entered for at least one ocular surface.

Name	Required?	Usage
TOTAL EXPOSURE TIME (s)	Yes	Enter the total time a light source is irradiating the eye in a continuous, pulsed, or scanning manner.
SPECTRUM	Yes	Either enter wavelengths (first column) and corresponding spectral intensity values (second column) in the table, or press "Load" to select a two-column numeric delimited text file with wavelengths and corresponding spectral intensities.
STATIC BEAM RADIOMETRIC MEASUREMENTS	Yes	The rows of both columns requiring values are based on the source characteristics and the row-specific parameters of ISO 15004-2:2024 Tables 2-6 (irradiated ocular surface, wavelength range, exposure duration). Values entered in other rows are automatically deleted. First column contains aperture diameters used during power/energy measurements. If "from standard" is selected in the column header, the aperture diameters are automatically entered by the app based on the source parameters and ISO 15004-2:2024 Table 6. If "user-entered" is selected, the user may specify alternate aperture diameters used during power/energy measurements. Second column contains values for the radiometric measurement type selected in the column header: average power, cumulative energy for the total exposure time, or single-pulse energy (only if "SOURCE TYPE" is pulsed). If the source spectrum has been normalized to an area of 1, enter the power or energy through each corresponding aperture integrated across the entire spectrum. If the spectrum is not normalized to an area of 1, enter the scaling factor to yield an estimate of the integrated power or energy through each corresponding aperture.
Pulsed source parameters	If "SOURCE TYPE" is pulsed	Enter the pulse duration (first cell) and either the total number of pulses emitted during the total exposure time or the pulse repetition frequency (second cell).
Scanning beam parameters	If "SOURCE TYPE" is scanning	For each ocular surface, in the first row enter either the scan pulse duration or the scan velocity, and in the second row enter either the number of scan pulses during the total exposure time or the scan pulse repetition frequency. If an ocular surface is not irradiated or the beam is static on that surface, enter zero for the scan pulse duration or scan velocity, and the number of scan pulses or scan pulse frequency is ignored.

4. Outputs

Name	Description
Relevant Rows of Tables 2-6	Titles appear in the rows corresponding to ISO 15004-2:2024 Tables 2-6 which match the source characteristics (irradiated ocular surface, wavelength range, exposure duration).
Effective exposure	Total and single pulse (only for pulsed and scanning sources) weighted radiant exposure values for the source are calculated for each relevant row based on the corresponding spectral and temporal characteristics.
Group 1 Limits	Total and single pulse (only for pulsed and scanning sources) exposure limits for each relevant row of ISO 15004-2:2024 Tables 2 and 3.
Group 2 RMEs	Total and single pulse (only for pulsed and scanning sources) recommended maximum exposures for each relevant row of ISO 15004-2:2024 Tables 4 and 5.
GROUP 1 HAZARDS	Each hazard value is the ratio of an effective exposure to corresponding Group 1 limit, and the largest hazard value is displayed (whether from total or single pulse exposure) for each of the two ocular regions (anterior: cornea/lens and iris; posterior: retina) and each of the two photodamage mechanisms (photochemical, photothermal).
GROUP 2 HAZARDS	Each hazard value is the ratio of an effective exposure to corresponding Group 2 RME, and the largest hazard value is displayed (whether from total or single pulse exposure) for each of the two ocular regions (anterior: cornea/lens and iris; posterior: retina) and each of the two photodamage mechanisms (photochemical, photothermal).
MULTIPLE SOURCE HAZARDS	For a device with multiple sources, all values in each source's GROUP 1 HAZARDS table are added across the sources per ocular region and photodamage mechanism. For the values in each source's GROUP 2 HAZARDS table, the photochemical hazards are added across the sources per ocular region, but the photothermal hazards are added together only for those sources irradiating simultaneously (as specified with "Simultaneous with next source #"). The Group 2 photothermal hazards shown here always show the maximum hazards for the device, whether from a single source or combined sources.

5. Software app setup

System requirements: 64-bit Windows 10/11, 10 GB free hard drive space

Required pre-installed software:

Software component	URL
MATLAB Runtime R2024a (24.1) for 64-bit Windows	https://ssd.mathworks.com/supportfiles/downloads/R2024a/Release/9/deployment_files/installer/complete/win64/MATLAB_Runtime_R2024a_Update_9_win64.zip
LabVIEW Runtime 2026 Q1 for 64-bit Windows	https://www.ni.com/en/support/downloads/software-products/download.labview-runtime.html#585815

Unzip the file **OLHcalc_v005.zip** to any full-access folder in Windows.

The main app file **OLHcalc_v005.exe** will be placed in this folder, and a subfolder named **data** will be created with the files **ISO15004_9Native.dll** and **ISO15004_multisourcehazardNative.dll**.

6. Basic Operating Steps

The app updates calculations automatically about every **500 ms**.

Step 1 — Set global options

Optionally enter **Device name**.

Check the box for these as applicable to the device and desired method of hazard assessment:

- **Phakic**
- **Immobilized eye**

Step 2 — Enter characteristics for each source

In **Source #**, add/increment one page for each optical source, starting with zero.

Enter information in the following recommended order:

- **Source name** (optional),
- **SOURCE TYPE**
- **TOTAL EXPOSURE TIME (s)**
- **STATIC BEAM DIAMETER (mm)**
- **TOTAL EXPOSURE TIME (s)**
- **SPECTRUM**, entered or loaded from a file
 - ⚠ **Use two-column numeric delimited text files only.**
- **Pulsed source parameters**, if applicable
- **Scanning beam parameters**, if applicable

Step 3 — Check Relevant Rows of Tables 2-6

Step 4 — Enter STATIC BEAM RADIOMETRIC MEASUREMENTS

Select whether aperture diameters are **from standard** or **user-entered**.

Select measurement type of **Cumulative energy (J)**, **Average power (W)**, or **Single pulse energy (J)** (allowed only for static pulsed sources).

Step 5 — Define simultaneous sources for multiple source devices

Use **Simultaneous with next source #** to identify which sources irradiate at the same time.

 Delete any unneeded sources with the **Delete source** button.

Step 6 — Review results

Check:

- **Effective exposure**
- **Group 1 Limits**
- **Group 2 Limits**
- **GROUP 1 HAZARDS**
- **GROUP 2 HAZARDS**
- **MULTIPLE SOURCE HAZARDS**

Step 7 — Save results

Increment through the sources in the device and use the Windows Snipping Tool or Print Screen keyboard button to save a window-based screenshot for each source.

Step 8 — Exit the app

Press the **EXIT** button when finished.

6. Troubleshooting

Symptom	Check
No useful result	Verify source type, measurement type, spectrum, and exposure inputs
Spectrum does not load	Confirm the file is numeric and delimited correctly
Combined hazard looks wrong	Review Simultaneous with next source # and source order

7. Good Operating Practices

Practice	Why it helps
Add one source at a time	Simplifies validation
Verify units before calculation	Reduces bad inputs
Recheck simultaneous-source mapping after edits	Prevents wrong combined hazards
Use clean numeric spectrum files	Reduces file-read failures

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